



UNITED INTERNATIONAL UNIVERSITY (UIU)
Dept. of Electrical & Electronic Engineering (EEE)
Final Project, Trimester: Summer, Year: 2026
Course: EEE 3310 (Sec: A, B)
Title: Digital Signal Processing Laboratory
Full Marks: **40**

Guidelines for completing the assigned project:

1. Carefully read the Project Assignment Sheet and identify the different components of the projects and their required outcomes.
2. The project involves the following key components:
 - a. Channel estimation: Estimation of the paths (direct path and reflected paths) and their delays and gains. You will need the channel sounding input and output data files and come up with a technique to estimate the channel delays and gains.
 - b. Channel filter: Once the delays and gains are estimated, form the filter transfer function $H(z) = B(z)/A(z)$. The parameters in $A(z)$ and $B(z)$ should be based on the estimated delays and gains from the channel estimation phase.
 - c. Equalizer design: This design of the equalizer is basically to undo the channel echo or multipath effect. This is done by using, for example, a zero-forcing equalizer. Do a basic study of the operation and implementation of this type of equalizer and implement it in this project.
 - d. Once you have come up with an equalizer, pass the audio file given to you and hear the sound at the output. The output audio file should have the echoes removed and should present a clear audio file. The quality of audio output, i.e., the removal of the echoes will depend upon the correct design of the equalizer.